SPOT on EM 1 in EM blow out.

RIVER HABITAT SURVEY 2003 Version

A FIELD SURVEY DETAILS

MACROPHYTES 6 – 8 spot checks

leave blank if new site

Site Number:

Site Reference: HAREMOPE ONE

Spot-check 1 Grid Ref: NU 05969 21475 (+10m)

Spot-check 6 Grid Ref: NU 06130 21303

End of site Grid Ref: NU 06250 21144

Reach Reference:

River name: TILL

Date 9 / 8 /2022 Time: 11:00

Surveyor name: Russ

Accredited Surveyor code:

Is the site part of a river or an artificial channel? River ☑ Artificial ☐

Are adverse conditions affecting survey? No ☑ Yes ☐

If yes, state .....

Is bed of river visible? barely or not ☐ partially ☐ ±entirely ☑

Is health and safety assessment form attached? Yes ☑ No ☐

Number of photographs taken: 2

Photo references:

Site surveyed from: left bank ☑ right bank ☑ channel ☑

☐ **When options shown with 'shadow boxes', tick one box only**

LEFT banks determined by facing downstream RIGHT

B PREDOMINANT VALLEY FORM (within the horizon limit) (tick one box only)

(tick one box only)



☐ shallow vee



☐ deep vee



☐ gorge



☑ concave/bowl



☐ asymmetrical valley



☐ U-shape valley



☐ no obvious valley sides

Distinct flat valley bottom? No ☐ Yes ☑

Natural terraces? No ☑ Yes ☐

C NUMBER OF RIFFLES, POOLS AND POINT BARS (enter total number in boxes)

Riffle(s)	5	Unvegetated point bar(s)	0
Pool(s)	3	Vegetated point bar(s)	0

D ARTIFICIAL FEATURES (indicate total number of occurrences of each category within the 500m site)

If none, tick box ☑

	Major	Intermediate	Minor		Major	Intermediate	Minor
Weirs/sluices				Outfalls/ intakes			
Culverts				Fords			
Bridges				Deflectors/ groynes/croys			
Other - state							

Is channel obviously realigned? No ☐ Yes, <33% of site ☐ ≥33% of site ☐

Is channel obviously over-deepened? No ☑ Yes, <33% of site ☐ ≥33% of site ☑

Is water impounded by weir/dam? No ☑ Yes, <33% of site ☐ ≥33% of site ☐

SITE REF.	**RIVER HABITAT SURVEY: TEN SPOT-CHECKS**	**Page 2 of 4**

Spot-check 1 is at: upstream end ☐ downstream end ☑ of site (tick one box)

E PHYSICAL ATTRIBUTES (to be assessed across channel within 1m wide transect)

When boxes 'bordered', only one entry allowed	1 GPS	2	3	4	5	6 GPS	7	8	9	10	GPS
LEFT BANK	Ring EC or SC if composed of sandy substrate										
Material NV, BE, BO, CO, GS, EA, PE, CL, CC, SP, WP, GA, BR, RR, TD, FA, BI	SA	SA	SA	EA	EA	EA	EA	SA	SA	SA	
Bank modification(s) NK, NO, RS, RI, PC(B), BM, EM	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	
Marginal & bank feature(s) NV, NO, EC, SC, PB, VP, SB, VS, NB	SC	EC	SC	SC	NO	EC	NO	NB	NB	NO	
CHANNEL	GP- ring either G or P if predominant										
Channel substrate NV, BE, BO, CO, GP, SA, SI, CL, PE, EA, AR	GP	SA	GP	GP	GP	GP	GP	CO	CO	GP	SI
Flow-type NV, FF, CH, BW, UW, CF, RP, UP, SM, NP, DR	SM	SM	SM	SM	UW	SM	SM	SM	RP	SM	
Channel modification(s) NK, NO, CV, RS, RI, DA, FO	RS	RS	RS	RS	RS	RS	RS	RS	RS	RS	
Channel feature(s) NV, NO, EB, RO, VR, MB, VB, MI, TR	NO	NO	NO	NO	NO	NO	NO	NO	RO	NO	
For braided rivers only: number of sub-channels					NA						
RIGHT BANK	Ring EC or SC if composed of sandy substrate										
Material NV, BE, BO, CO, GS, EA, PE, CL, CC, SP, WP, GA, BR, RR, TD, FA, BI	EA	EA	SA	SA	EA	EA	EA	EA	SA	SA	
Bank modification(s) NK, NO, RS, RI, PC(B), BM, EM	NO	NO	NO	SC	NO	NO	NO	NO	NO	NO	
Marginal & bank feature(s) NV, NO, EC, SC, PB, VP, SB, VS, NB	NO	NO	SC	NO	SC	NO	EC	SC	NT	SB	

Enter channel substrate(s) not occurring as predominant in spot-checks but present in >1% of whole site.

F BANKTOP LAND-USE AND VEGETATION STRUCTURE (to be assessed over a 10m wide transect)

Land-use: choose one from BL, BP, CW, CP, SH, OR, WL, MH, AW, OW, RP, IG, TH, RD, SU, TL, IL, PG, NV

		1	2	3	4	5	6	7	8	9	10
LAND-USE WITHIN 5m OF LEFT BANKTOP		SH	TH	TH	TH	TH	TH	SU	TH	SH	TH
LEFT BANKTOP (structure within 1m)	B/U/S/C/NV	S	U	U	S	U	S	S	U	S	S
LEFT BANK-FACE (structure)	B/U/S/C/NV	S	S	U	S	U	S	S	S	S	U
RIGHT BANK-FACE (structure)	B/U/S/C/NV	S	S	S	S	S	S	S	S	S	S
RIGHT BANKTOP (structure within 1m)	B/U/S/C/NV	S	S	U	S	S	S	S	U	S	S
LAND-USE WITHIN 5m OF RIGHT BANKTOP		TH	TH	TH	SH	TH	TH	TH	TH	BL	BL

G CHANNEL VEGETATION TYPES (to be assessed over a 10m wide transect: use E (≥ 33% area), ✓ (present) or NV (not visible)

	1	2	3	4	5	6	7	8	9	10	
None (✓) or Not Visible (NV)											
Liverworts/mosses/lichens											
Emergent broad-leaved herbs		✓	✓	✓	✓	✓	✓		✓	✓	✓
Emergent reeds/sedges/rushes/grasses/horsetails		E	✓	✓	✓	✓	✓	✓	E	✓	✓
Floating-leaved (rooted)				NV							
Free-floating											
Amphibious	✓						✓	✓	✓	✓	✓
Submerged broad-leaved										✓	
Submerged linear-leaved											
Submerged fine-leaved											
Filamentous algae	✓	E	E	✓	✓	✓	✓	E	✓	E	✓

Use end column for overall assessment over 500m, including types not occurring in spot-checks (use ✓, E or NV)

SITE REF.	**RIVER HABITAT SURVEY : 500m SWEEP-UP**	**Page 3 of 4**

H LAND-USE WITHIN 50m OF BANKTOP

Use ✓ (present) or E (≥ 33% banklength)

	L	R		L	R
Broadleaf/mixed woodland (semi-natural) (BL)			Natural open water (OW)		
Broadleaf/mixed plantation (BP)			Rough/unimproved grassland/pasture (RP)		
Coniferous woodland (semi-natural) (CW)			Improved/semi-improved grassland (IG)		
Coniferous plantation (CP)			Tall herb/rank vegetation (TH)		✓
Scrub & shrubs (SH)	✓	✓	Rock, scree or sand dunes (RD)		
Orchard (OR)			Suburban/urban development (SU)		
Wetland (e.g. bog, marsh, fen) (WL)			Tilled land (TL)	✓	
Moorland/heath (MH)			Irrigated land (IL)		
Artificial open water (AW)			Parkland or gardens (PG)		
			Not visible (NV)		

I BANK PROFILES

Use ✓ (present) or E (≥ 33% banklength)

Natural/unmodified	L	R	**Artificial/modified**	L	R
Vertical/undercut	✓	✓	Resectioned (reprofiled)		
Vertical with toe	✓	✓	Reinforced - whole		
Steep (>45°)	E	E	Reinforced - top only		
Gentle	✓	✓	Reinforced - toe only		
Composite			Artificial two-stage		
Natural berm	✓	✓	Poached bank		
			Embanked		
100% both banks ←			Set-back embankment		

J EXTENT OF TREES AND ASSOCIATED FEATURES

*record even if <1%

TREES (tick one box per bank)

	Left	Right
None	☑	☑
Isolated/scattered	☐	☐
Regularly spaced, single	☐	☐
Occasional clumps	☐	☐
Semi-continuous	☐	☐
Continuous	☐	☐

ASSOCIATED FEATURES (tick one box per feature)

	None	Present	E (≥33%)
Shading of channel	☐	☑	☐
*Overhanging boughs	☐	☑	☐
*Exposed bankside roots	☑	☐	☐
*Underwater tree roots	☑	☐	☐
Fallen trees	☑	☐	☐
Large woody debris	☑	☐	☐

K EXTENT OF CHANNEL AND BANK FEATURES

(tick one box for each feature) *record even if <1%

	None	Present	E(≥33%)
*Free fall flow	☑	☐	☐
Chute flow	☑	☐	☐
Broken standing waves	☑	☐	☐
Unbroken standing waves	☐	☑	☐
Rippled flow	☐	☑	☐
*Upwelling	☐	☑	☐
Smooth flow	☐	☐	☑
No perceptible flow	☐	☑	☐
No flow (dry)	☑	☐	☐
Marginal deadwater	☐	☑	☐
Eroding cliff(s)	☐	☑	☐
Stable cliff(s)	☐	☑	☐

	None	Present	E (≥33%)
Exposed bedrock	☑	☐	☐
Exposed boulders	☑	☐	☐
Vegetated bedrock/boulders	☑	☐	☐
Unvegetated mid-channel bar(s)	☑	☐	☐
Vegetated mid-channel bar(s)	☑	☐	☐
Mature island(s)	☑	☐	☐
Unvegetated side bar(s)	☐	☑	☐
Vegetated side bar(s)	☑	☐	☐
Unvegetated point bar(s)	☑	☐	☐
Vegetated point bar(s)	☑	☐	☐
*Unvegetated silt deposit(s)	☐	☐	☐
*Discrete unvegetated sand deposit(s)	☑	☐	
*Discrete unvegetated gravel deposit(s)	☑	☐	

SITE REF.	**RIVER HABITAT SURVEY : DIMENSIONS AND INFLUENCES**	**Page 4 of 4**

L CHANNEL DIMENSIONS (to be measured at one location on a straight uniform section, preferably across a riffle)

LEFT BANK		CHANNEL		RIGHT BANK	
Banktop height (m)	2.5	Bankfull width (m)		Banktop height (m)	2.5
Is banktop height also bankfull height? (Y or N)	Y	Water width (m)	10	Is banktop height also bankfull height? (Y or N)	
Embanked height (m)	(set back)	Water depth (m)	0.2	Embanked height (m)	(set back)

If trashline (or notch) lower than banktop, indicate: height above water (m) = X width from bank to bank (m) = X

Bed material at site is: consolidated ☐ unconsolidated (loose) ☑ unknown ☐

Location of measurements is: riffle ☑ other ☐ (state) SC 6

M FEATURES OF SPECIAL INTEREST Use ✓ or E (≥ 33% length) *record even if <1%

None	☐	Very large boulders (>1m)	☐	Backwater(s)	☑	Marsh(es)	☐
Braided channels	☐	*Debris dam(s)	☐	Floodplain boulder deposits	☐	Flush(es)	☐
Side channel(s)	☐	*Leafy debris	☐	Water meadow(s)	☐	Natural open water	☐
*Natural waterfall(s) > 5m high	☐	Fringing reed-bank(s)	☐	Fen(s)	☐	Others (state)	☐
*Natural waterfall(s) < 5m high	☐	Quaking bank(s)	☐	Bog(s)	☐		
Natural cascade(s)	☐	*Sink hole(s)	☐	Wet woodland(s)	☐		

N CHOKED CHANNEL (tick one box)

Is 33% or more of the channel choked with vegetation? No ☑ Yes ☐

O NOTABLE NUISANCE PLANT SPECIES Use ✓ or E (≥ 33% length) *record even if <1%

		bankface	banktop to 50m		bankface	banktop to 50m
None ☑	*Giant hogweed	☐	☐	*Himalayan balsam	☐	☐
	*Japanese knotweed	☐	☐	*Other (state).....	☐	☐

P OVERALL CHARACTERISTICS (Circle appropriate words, add others as necessary)

Major impacts: landfill - tipping - litter - sewage - pollution - drought - abstraction - mill - dam - road - rail - industry - housing mining - quarrying - overdeepening - afforestation - fisheries management - silting - waterlogging - hydroelectric power

Evidence of recent management: dredging - bank mowing - weed cutting - enhancement - river rehabilitation - gravel extraction - other (please specify)

Animals: otter - mink - water vole - kingfisher - dipper - grey wagtail - sand martin - heron - dragonflies/damselflies

Other significant observations: if necessary use separate sheet to describe overall characteristics and relevant observations

canadian pondweed

Q ALDERS (tick one box in each of the two categories) *record even if <1%

*Alders? None ☐ Present ☑ Extensive ☐

*Diseased Alders? None ☑ Present ☐ Extensive ☐

R FIELD SURVEY QUALITY CONTROL (✓ boxes to confirm checks)

Have you taken at least two photos that illustrate the general character of the site and additional photos of any weirs/ sluices and major/intermediate structures across the channel?	☑
Have you completed all ten spot-checks and made entries in all boxes in E & F on page 2?	☑
Have you completed column 11 of section G (and E if appropriate) on page 2?	☑
Have you recorded in section C the number of riffles, pools and point bars (even if 0) on page 1?	☑
Have you given an accurate (alphanumeric) grid reference for spot-checks 1, 6 and end of site (page 1)?	☑
Have you stated whether spot-check 1 is at the upstream or downstream end of the site (top of page 2)?	☑
Have you cross-checked your spot-check and sweep-up responses with the channel modification indicators given on page 2 of the spot-check key?	☐